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Modeling Demographic Variation in A&E Attendance Risk Across Scotland

by

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Executive Summary

Accident and Emergency (A&E) departments across the UK, including Scotland, are under pressure from high attendance which has been associated with longer wait times, increased costs, and poor patient outcomes. Recognizing who is most likely to attend is important for understanding how to manage that pressure. Our research investigates who is most likely to attend A&E in Scotland based on the individuals age, sex, level of socioeconomic deprivation, and health board treatment area.

Using monthly A&E attendance records (2021-2026) published by Public Health Scotland, combined with yearly population estimates from National Health Scotland and health board deprivation data from the Scottish Index of Multiple Deprivation (SIMD), our research built a statistical model of attendance risk that accounts for how many people are in each subpopulation. Our analysis found that no single factor on its own can explain who is most at risk to attend A&E. Age, deprivation, and health board area each matter significantly and they interact with another. For instance, the effect deprivation has on attendance risk is not the same for every age group nor is it the same across Scotland. Risk was greater among the youngest (under 18-24) and the oldest (75 plus), deprivations effect on attendance was greatest in those of working age (25-64), in some health board areas the demographic factors compounded to produce much higher attendance risks, and sex had very minor effect. It was found that the attendance risk of the subpopulation with the highest modeled risk was nearly 10 times bigger than the subpopulation with the lowest modeled risk.

These findings suggest to reduce avoidable attendance and, in turn, pressure on Scotland's A&E departments, strategies aimed at specific high risk group combinations of age, deprivation, and health are more likely to be effective. This provides Public Health Scotland and health boards with a starting point backed by evidence to help potentially implement alternative care services towards subpopulations at most risk and find ways to prioritize resources. Future research could extend this work by analyzing specific indicators of deprivation, frequent/repeat attendee population, and effects of the COVID-19 pandemic.

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Generative AI Disclosure

Project Studies: Generative AI was used in the following ways in the project:

- Claude, assisted in *minor* code edits and errors

Proofing: Generative AI was used to assist in checking for potential grammar errors.

- Claude, August 15, Grammar check

Contents

1	Introduction	1
2	Data Sources and Preprocessing	1
2.1	Overview	1
2.2	Preprocessing	1
2.3	COVID-19 Outlier Exclusion	2
3	Exploratory Data Analysis	2
4	Methodology	5
4.1	Baseline Model	6
4.2	Saturated Fixed Effects Model	6
4.3	Saturated Interaction Model	6
4.4	Model Comparison	6
5	Results	7
5.1	Factor and Interaction Term Importance	8
5.2	Modeled Rates	9
5.3	Subpopulation Risk	13
6	Discussion	13
6.1	Key Findings	13
6.2	Future Research	14
6.3	Limitations	14
7	Conclusion	15
	Appendices	17
A	Health Board Name and Country Code	17
B	Model Coefficients	17

1 Introduction

Accident and Emergency (A&E) departments in the UK have been experiencing increases in attendance which has been associated with longer wait times, increased costs, and crowding [14]. Crowding and long wait times themselves have been connected to increased mortality, with past research finding an association between prolonged wait times in A&E and potentially avoidable deaths [5]. Understanding what drives A&E demand is not only a matter of operational efficiency but also patient safety.

Public Health Scotland (PHS), has been developing a Whole System Model intended to simulate patient flow across A&E departments and support health boards and the Scottish government in planning for future demand. Publicly available datasets from Public Health Scotland support this work by providing evidence such as monthly attendance counts, demographics of attendances, and SIMD (Scottish Index of Multiple Deprivation). This paper attempts to address the process of finding the drivers of A&E attendance.

The goal of our research is to bring attention to the demographic composition of A&E attendance to provide health boards and the Scottish Government with a basis for interventions and resource allocation towards the groups and areas facing the greatest pressure. The questions we aim to answer is, can we identify who is most at risk among A&E attendance in Scotland and to what extent do they vary across age, sex, deprivation, and health boards of treatment?

The paper is structured as follows. Section 2 describes the data sources and preprocessing done. Section 3 presents exploratory analysis. Section 4 sets out the statistical methodology and modeling approach. Section 5 presents results from our modeling, and Section 6 discusses their implications, limitations, and directions for future work.

2 Data Sources and Preprocessing

2.1 Overview

This research uses three publicly available datasets summarized in Table 1.

Dataset	Source	Use / Purpose
Monthly A&E Attendance by Demographics [11]	Public Health Scotland	Attendance counts by Health Board, Age, Sex, and Deprivation. (2018-2026)
Data Zone Population Estimates [9]	NHS Scotland	Population estimates by Age, Sex, and Data Zone for attendance rate calculations. (2001-2024)
SIMD 2020v2 (Population-Weighted) [12]	Public Health Scotland	Data zone to Health Board and Deprivation. quintile lookup. (2020)

Table 1: Datasets used, their sources, and what they were used for.

2.2 Preprocessing

Rows with an unrecorded Age, Sex, or Deprivation category were excluded from the monthly attendance dataset. Department Type (Type 1 and Type 3) were combined into a single attendance count throughout since our research was focused on overall attendance risk rather than care pathway. To obtain the population denominators required for calculating attendance rate, since SIMD (Scottish Index of Multiple Deprivation) quintiles are assigned at the data zone level, data zone populations estimates [9] were aggregated up to health board level and joined to the SIMD quintiles via the SIMD lookup file [12]. For each year of the population estimate data, age was grouped into the same six age bands used in the monthly attendance data. The Data Zone population estimates data spans from 2001 to 2024, however the monthly attendance data [11] spans into 2026. We acknowledge that while other population estimate data sets exist for 2025 through 2026, they did not contain the granularity required for our research. Additionally, it was found that the population growth from 2024 to 2026

was not significant enough to warrant building a model to predict population estimates for 2025-2026. Therefore, the most recent population estimates of 2024 were carried forward to cover the remaining months of 2025 and 2026. Monthly attendance and population estimate data was merged by Year, Health Board, Age, Sex, and Deprivation. The rate is calculated by dividing attendance counts for each grouping of Year, Health Board, Age, Sex, and Deprivation by their corresponding population figures. The rate is expressed as attendances per 1,000 population.

2.3 COVID-19 Outlier Exclusion

A&E attendance in Scotland dropped sharply during the COVID-19 pandemic, with the population attendance rate visualized in Figure 1. Given the research focus on estimating demographic related attendance risk and the limited scope available to properly model the pandemic disruption, the analysis was restricted to the period following the return of attendance rates close to pre-pandemic levels. The cutoff data was determined using monthly attendance rate expressed as a percentage of the 2019 baseline. Dates from June 2021 onward found attendance rates returned to 95% of the 2019 baseline. All earlier data (2018 to May 2021) was excluded. We note that this decision cuts down the number of months available for analysis. We also acknowledge the future research potential in analyzing the effect that the COVID-19 pandemic has had on A&E attendance.

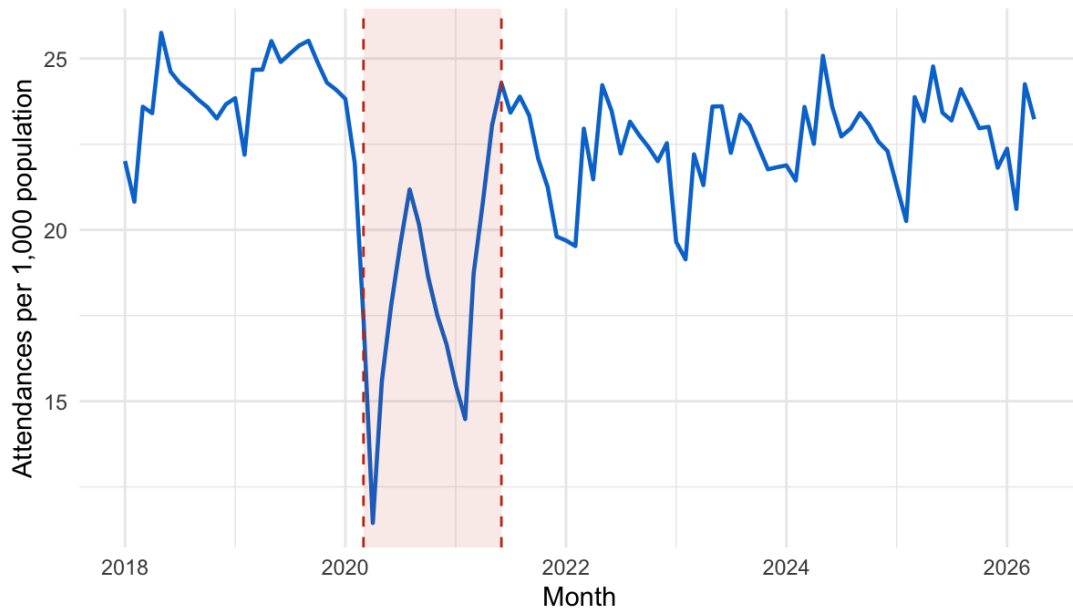


Figure 1: Monthly A&E attendances per 1,000 population for 2018-2026 with shaded region marking COVID-19 onset (March 2020) to return to 95% of pre-pandemic (2019) baseline (June 2021).

3 Exploratory Data Analysis

To motivate modeling decisions we work to identify which demographics show a plausible relationship to A&E attendance. Population attendance data was examined for each demographic of Age (Under 18, 18-24, 25-39, 40-64, 75 plus), Sex (Female, Male), and Deprivation (SIMD Quintiles), as well as by Health Boards. Note that deprivation quintiles are ranked from 1 being the most deprived and 5 being the least deprived. Population attendance is expressed as a population adjusted rate (attendances per 1,000 population) as outlined in Section 2. Note Appendix A provides a table of NHS Scotland Health Board geography codes and their corresponding names for reference.

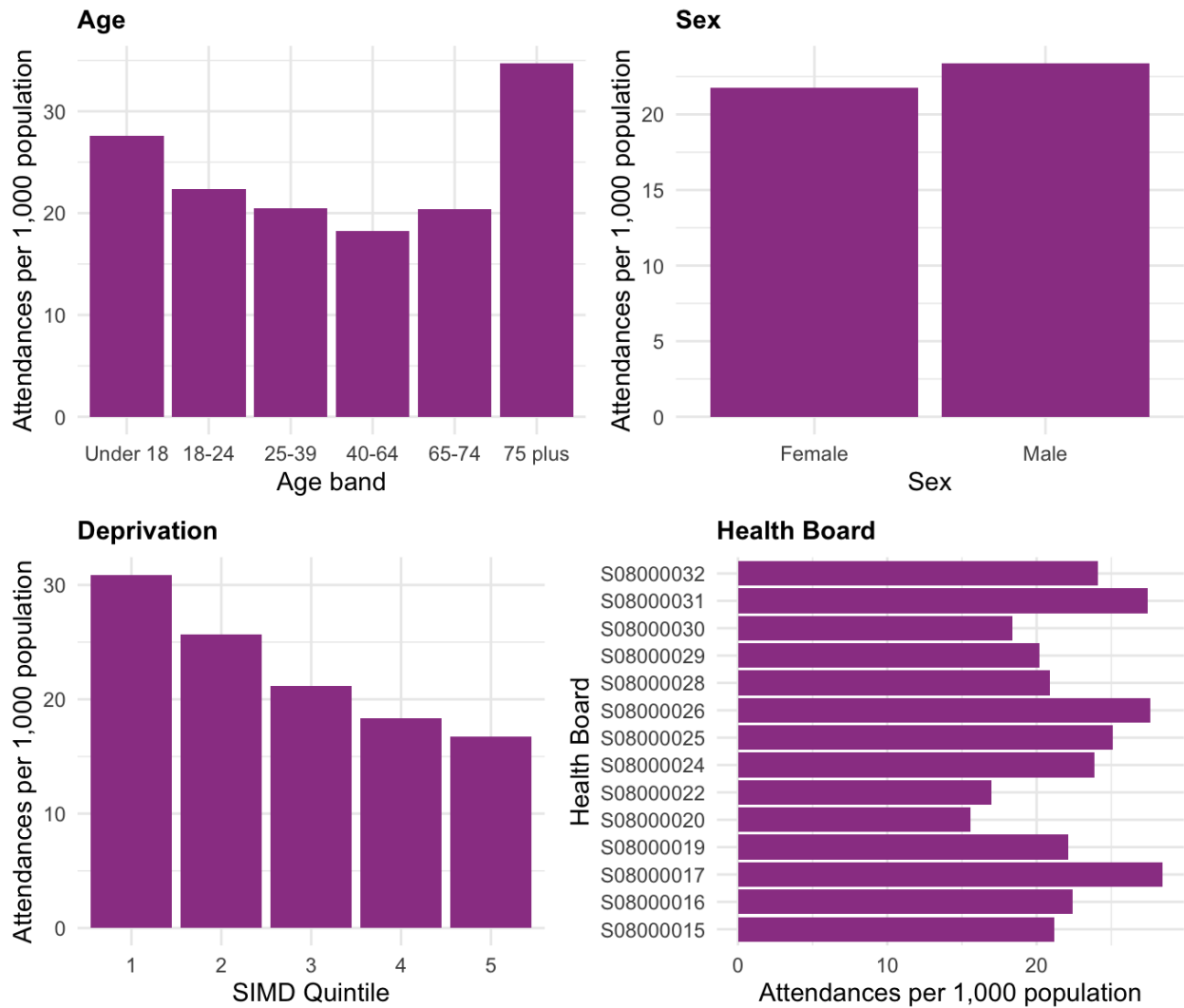


Figure 2: Population attendance rates expressed as attendances per 1,000 population stratified by age, sex, deprivation (SIMD quintile), and by health board.

We first consider possible main effect predictors of A&E attendance. Figure 2 presents the attendances per 1,000 population split by age, sex, deprivation quintiles, and health board of treatment. Across all four, there is clear divergence between the subgroups within, suggesting that A&E attendance is potentially effected by these factors. For instance, attendance rate was highest among the youngest (under 18) and oldest (75 plus) age bands at 27 and 34 per 1,000 population respectively and lowest among the 40-64 at just 18 per 1,000 population. Males showed a slightly higher attendance rate at 23 versus 21 per 1,000 population. Attendance rate increased with increasing deprivation from 16 per 1,000 in Quintile 5 (least deprived) to 30 per 1,000 population in Quintile 1 (most deprived). Attendance rate varied greatly by health board of treatment, ranging from 15 per 1,000 in S08000020 (NHS Grampian) to 28 per 1,000 in S08000017 (NHS Dumfries and Galloway) indicating geographic variation not explained by the demographic factors alone. Together, the visual evidence provides motivation to include age, sex, deprivation, and health board as predictors in our models.

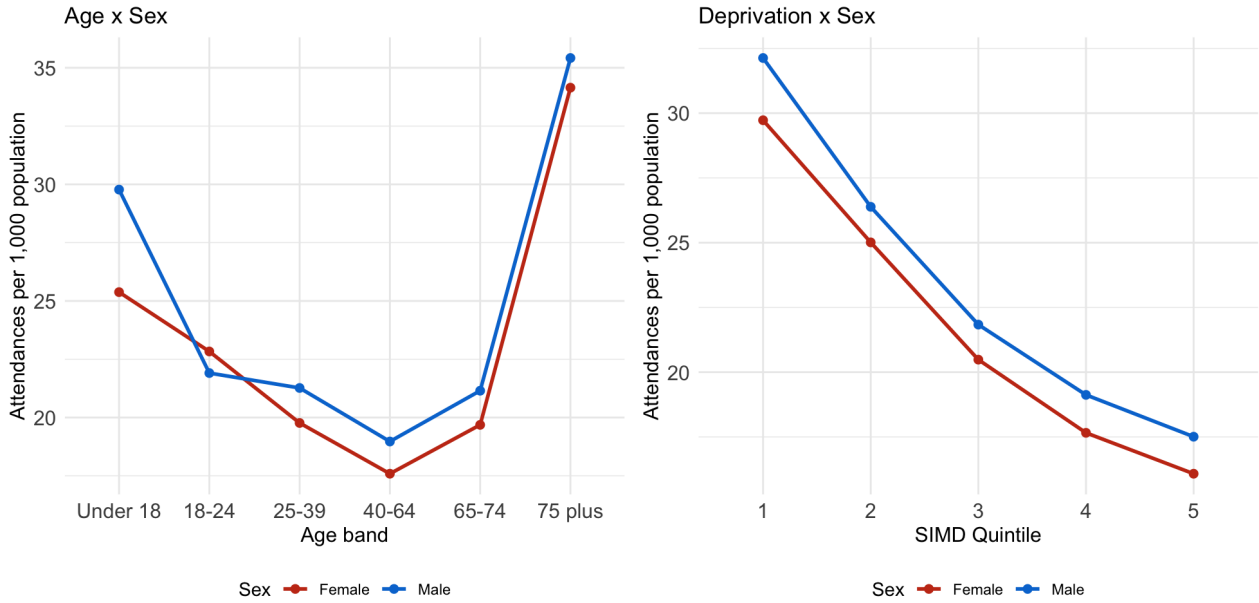


Figure 3: Population attendance rates as attendances per 1,000 population for two-way interactions of Age $\times$ Sex and Deprivation $\times$ Sex.

Figure 3 shows the interactions of Age $\times$ Sex and Deprivation $\times$ Sex. The male and female lines across age bands showed lines diverging and crossing in early age bands of Under 18 and 18-24 suggesting the sex gap in attendance rate was not consistent across age. The male and female lines remained mostly parallel across all deprivations quintiles with little visible divergence suggesting the effect of deprivation on attendance rate is broadly consistent across sex.

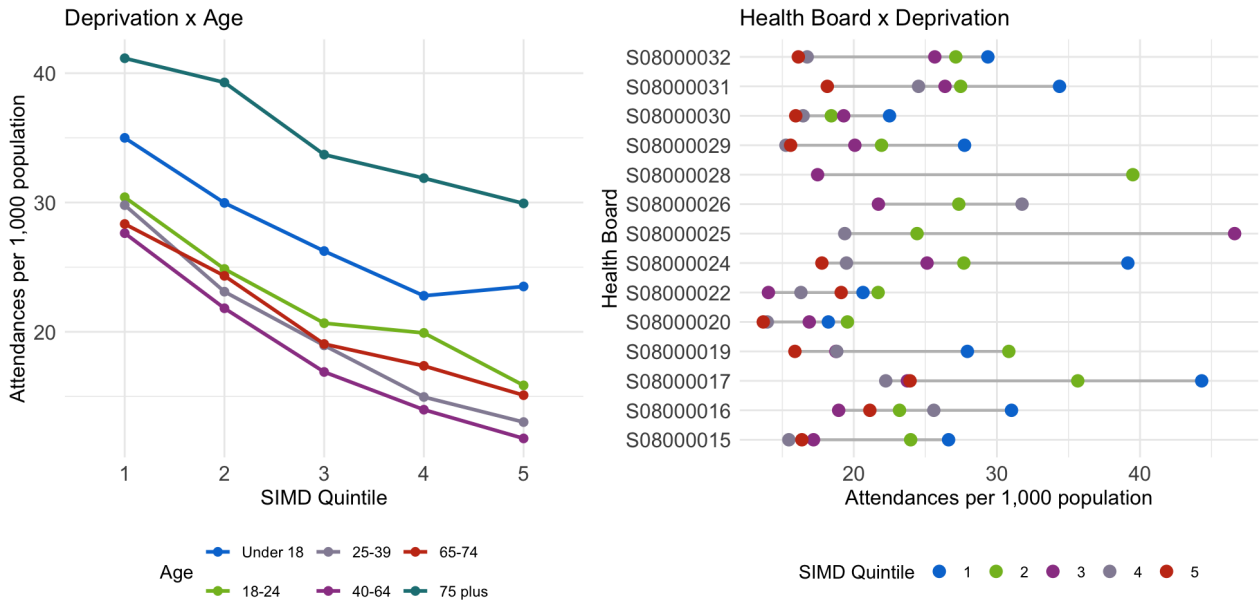


Figure 4: Population attendance rates as attendances per 1,000 population for two-way interactions of Deprivation $\times$ Age and Health Board $\times$ Deprivation.

Figure 4 shows the interactions of Age $\times$ Deprivation and Health Board $\times$ Deprivation. There appeared to be some divergence between age band lines across deprivation quintiles particularly between the most deprived (quintiles 1 and 2) and the least deprived (quintiles 4 and 5). For instance, Under 18 showed a steeper incline between Quintile 4 and 5 whereas 18-24 showed a steeper decline suggesting deprivation patterns in attendance rate differ with age. The horizontal spread between the deprivation

quintile points varied between health boards. For instance, S08000024 (NHS Lothian) saw the points much farther apart compared to S08000020 (NHS Grampian) where the deprivation points remained quite close together suggesting different relationships between deprivation and attendance rate between health boards.

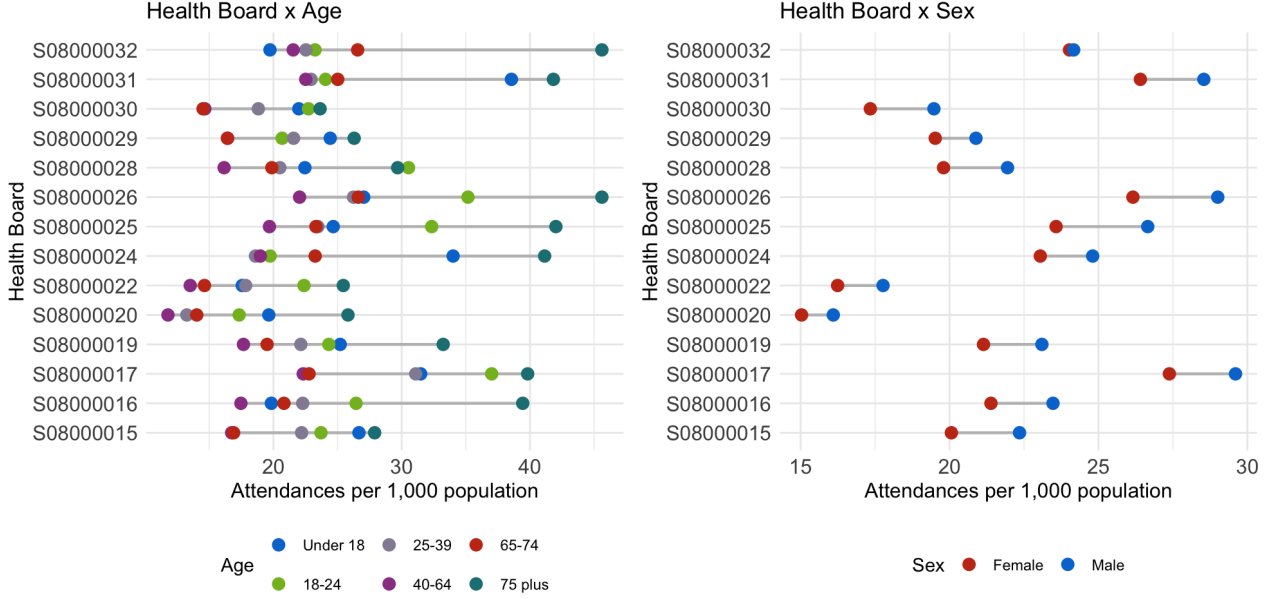


Figure 5: Population attendance rates as attendances per 1,000 population for two-way interactions of Health Board $\times$ Age and Health Board $\times$ Sex.

Figure 5 shows the interactions of Health Board $\times$ Age and Health Board $\times$ Sex. The horizontal ordering of Age bands across health boards was broadly similar with 75 plus sitting above the other age lines. However, in board S08000028 (NHS Western Isles) we see a different ordering with 18-24 being slightly higher. Across health boards we also see different spreads between age bands. The male and female points across health boards appear much more stable with little difference in the gender gap between health boards. We expect this to be a weaker interaction yet still worth testing in model building.

Overall, these explanatory patterns motivate testing of all six two-way interactions in Section 4.

4 Methodology

We aim to model the A&E attendance rate of the Scottish population given demographic information including Age, Sex, and Deprivation as well as Health Board of Treatment (HBT). Attendance counts were modeled using Generalized Linear Models (GLMs). A Poisson distribution was initially used, but preliminary testing identified significant overdispersion, a negative binomial distribution was therefore used throughout.

To convert modeled attendance *counts* into attendance *rates*, a population offset $\log(P_s)$, where P_s is the population of subpopulation s , (a specific combination of Age, Sex, Deprivation, and Health Board), was included in all models. This scaled the modeled counts proportionally with population size. So the attendance rate, depends only on the predictors, and is found by exponentiation of the sum of the relevant estimated coefficients. This is defined as

$$\frac{\mu_s}{P_s} = \exp(\text{linear predictors}),$$

where $\mu_s = E[A_s]$ is the expected attendance count for subpopulation s . This accounts for the underlying population size of each subpopulation, which is especially important given the range of population sizes across Sottish Health Boards.

A sequence of nested models of increasing complexity was fit, tested, and compared, defined as follows.

4.1 Baseline Model

The baseline model only included the intercept and the population offset, with no demographic predictors. It was defined as:

$$\log(\mu_{ijkl}) = \beta_0 + \log(P_{ijkl}),$$

where $\mu_{ijkl} = E[A_{ijkl}]$ is the expected number of attendances for age band i , sex j , deprivation quintile k , and Health Board ℓ where A_{ijkl} is the observed attendance count for that subpopulation. β_0 is the intercept and $\log(P_{ijkl})$ is the log-population offset for the same subpopulation. These definitions retain the same interpretation for all subsequent models.

4.2 Saturated Fixed Effects Model

Age, sex, deprivation, and health board were then added as fixed main effects. The saturated fixed effect model was defined as:

$$\log(\mu_{ijkl}) = \beta_0 + \beta_{\text{Age}[i]} + \beta_{\text{Sex}[j]} + \beta_{\text{Dep}[k]} + \beta_{\text{HBT}[\ell]} + \log(P_{ijkl}),$$

where $\beta_{\text{Age}[i]}$ is the coefficient for the i th level of Age (and analogously for Sex, Deprivation, and HBT) where Under 18, Female, Deprivation Quintile 1, and Health Board S08000015 are the reference levels. This model assumes each variables effect on attendance rate is constant across levels of every other variable. These definitions retain the same interpretation for all subsequent models.

4.3 Saturated Interaction Model

All six possible two-way interaction terms were then added to the fixed effects model. The saturated interaction model was defined as:

$$\begin{aligned} \log(\mu_{ijkl}) = & \beta_0 + \beta_{\text{Age}[i]} + \beta_{\text{Sex}[j]} + \beta_{\text{Dep}[k]} + \beta_{\text{HBT}[\ell]} + \beta_{\text{Age}[i]:\text{Sex}[j]} + \beta_{\text{Age}[i]:\text{Dep}[k]} \\ & + \beta_{\text{Age}[i]:\text{HBT}[\ell]} + \beta_{\text{Sex}[j]:\text{Dep}[k]} + \beta_{\text{Sex}[j]:\text{HBT}[\ell]} + \beta_{\text{Dep}[k]:\text{HBT}[\ell]} + \log(P_{ijkl}), \end{aligned}$$

where each interaction term $\beta_{\text{Age}[i]:\text{Sex}[j]}$ (and analogously for the remaining five interactions) is the coefficient for the combination of the i th level of Age and the j th level of Sex, relative to the reference combination. Note the reference factors are still Under 18, Female, Deprivation Quintile 1, and Health Board S08000015.

4.4 Model Comparison

Table 2 summarizes model fit across the three nested negative binomial GLM model specifications using AIC, BIC, and Likelihood Ratio Test (LRT).

Fit	Residual df	LRT	χ^2 p-value	AIC	BIC
Model 1: Baseline (offset only)	44,489			452,503.9	452,521.3
Model 2: Saturated Fixed Effect	44,466	41,847.7	p<0.001	410,702.2	410,919.8
Model 3: Saturated Interaction	44,314	36,678.2	p<0.001	374,328.0	375,868.4

Table 2: Comparison results of nested population attendance rate models including AIC, BIC, and likelihood ratio test (LRT).

AIC decreased monotonically across the models, from 452,503.9 for the baseline model to 410,702.2 with the addition of the four fixed effects, and to 374,328.0 with the addition of all six two-way interactions. BIC followed the same pattern. The additions of the fixed effects as well as the addition of the interactions produced a statistically significant improvement in fit (LRT $p < 0.0001$). Note that each likelihood ratio test in Table 2 compares models against the preceding model rather than the baseline. The addition of interaction terms produced the largest improvement in AIC, BIC and LRT. On this basis, the saturated interaction model was chosen as the final model.

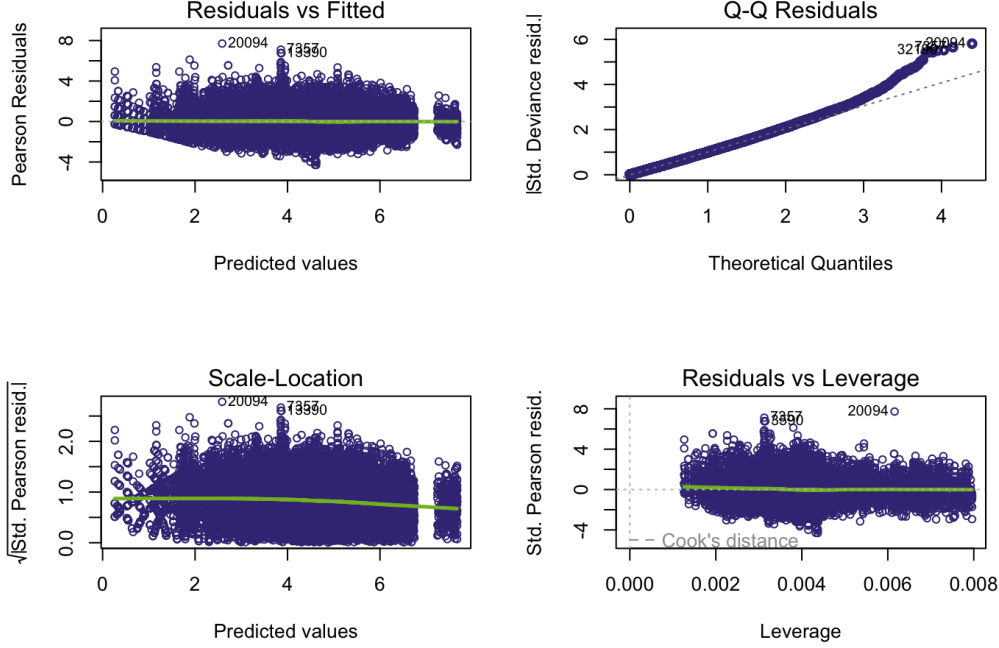


Figure 6: Diagnostic plots for the final chosen saturated interaction model including residuals versus fitted values, Q-Q plot, scale-location, and residuals versus leverage.

To Provide further support for the selected model, standard diagnostic checks were conducted for the final chosen model, Model 3, seen in Figure 6. The residuals versus fitted plot showed a smooth trend close to zero showing absence of significant heteroskedasticity. The QQ plot showed a slightly heavier upper tail suggesting a few observations with larger positive residuals. The scale-location plot showed a stable smooth trend. The residuals vs leverage plot showed uniformly low leverage and there is no indication that any observations had a disproportionate influence on the fitted model. Overall, the diagnostics show a model that is well-specified and little suggestion of problematic observations.

A limitation to note is that seven coefficients within the Deprivation $\times$ HBT interaction terms could not be estimated. That is, the particular Health Board Deprivation combinations of S08000025 (NHS Orkney) and S08000026 (NHS Shetland) in national SIMD Quintiles 4 and 5, and S08000028 (NHS Western) Isles in Quintiles 3, 4, and 5 had zero observed population, so no coefficient could be estimated for them. The model thus could not produce a predicted attendance rate for these specific non-existent Health Board Deprivation combinations. This is a limitation of the underlying population structure rather than of the modeling approach.

5 Results

Given the large number of coefficients in the final model ($n=183$) due to the number of factor levels and interaction terms, results are presented through factor and interaction importance, average population modeled rates, and a few named subpopulations of interest rather than discussion of individual

coefficients. For reference, the complete coefficient table has been provided in Appendix B.

5.1 Factor and Interaction Term Importance

Table 3 summarizes the change in model fit when each demographic factor (its main effect and all interactions involving it) was removed from the final model. We sought out which factors explained more of the variability in attendance risk, where factor removals with smaller changes indicated they explained comparatively little.

Factor Removed	AIC	Δ AIC	Residual df	LRT	χ^2 p-value
None	374,328		44,314		
Age	427,126	52,798	44,409	52,988.4	p<0.001
Sex	378,195	3,867	44,337	3,913.3	p<0.001
Deprivation	420,835	46,507	44,387	46,653.1	p<0.001
HBT	424,702	50,374	44,450	50,646.6	p<0.001

Table 3: Summary of change in model fit when each factor along with all its interactions are removed from the full model.

Age was found to be the most notable factor in the model, producing the largest change in fit when removed (Δ AIC = 52,798, LRT $\chi^2 = 52,988.4$, $p < 0.001$), followed closely by HBT (Δ AIC = 50,374, LRT $\chi^2 = 50,646.6$, $p < 0.001$) and Deprivation (Δ AIC = 46,507, LRT $\chi^2 = 46,653.1$, $p < 0.001$). Sex, in contrast, contributed the smallest change in fit when removed (Δ AIC = 3,867, LRT $\chi^2 = 3,913.3$, $p < 0.001$), indicating that although statistically significant, Sex explains comparatively little of the variability in attendance risk relative to Age, Deprivation, and Health Board. This ranking suggests, age, deprivation, and health boards are the primary drivers of variation in A&E attendance risk, with sex playing a minor role. As each factors removal also removes its interactions with the remaining three factors, these changes reflect each factors *total* contribution to the models explanatory power.

Interaction Removed	AIC	Δ AIC	df	LRT	χ^2 p-value
None	374,328				
Age:Sex	375,730	1,404	5	1,414.0	p<0.001
Age:Deprivation	381,960	7,634	20	7,673.7	p<0.001
Age:HBT	398,391	24,065	65	24,194.7	p<0.001
Sex:Deprivation	374,351	25	4	33.2	p<0.001
Sex:HBT	374,496	170	13	196.0	p<0.001
Deprivation:HBT	399,003	24,677	45	24,766.7	p<0.001

Table 4: Summary of change in model fit when individual interaction terms were removed from the full model.

Table 4 presents the change in model fit upon removal of each individual interaction term, holding all main effects and remaining interaction terms fixed. We considered how much the AIC decreased from the full interaction model and the LRT score to understand how much each interaction contributes to the model strength and thus how much improvement they each provide.

Deprivation $\times$ HBT was the found to have the greatest effect of an individual interaction (Δ AIC = 24,677, LRT $\chi^2 = 24,766.7$, $p < 0.001$), followed closely by Age $\times$ HBT (Δ AIC = 24,065, LRT $\chi^2 = 24,194.7$, $p < 0.001$). Age $\times$ Sex provided a small change (Δ AIC = 1,404, LRT $\chi^2 = 1,414.0$, $p < 0.001$), while Sex $\times$ HBT (Δ AIC = 170, LRT $\chi^2 = 196.0$, $p < 0.001$) and, in particular, Sex $\times$ Deprivation (Δ AIC = 25, LRT $\chi^2 = 33.2$, $p < 0.001$) saw a very little effect on the model. Thus, models with Sex interactions included only provided a modest improvement despite remaining statistically significant, in line with findings from Section 3. This suggests that health board in particular, in combination with deprivation and age, accounts for the majority of the interaction structure, while

sex's effect on attendance risk was stable but small across the other factors. The factor level results in Table 3 points to a consistent picture where attendance risk depends jointly on where a person receives treatment (health board), their age, and their level of deprivation, with these factors showing strong interactions between each other.

5.2 Modeled Rates

Having established which factors and interactions contribute most to explaining attendance risk, we now report on the modeled attendance rates to make these findings more concrete. For each level of Age, Sex, Deprivation, and Health Board, a modeled rate was calculated by taking the fitted attendance count (from the final chosen model) for every observed combination of factors containing that level, then dividing by that combinations population, then averaging across all those combinations. The average population rates help us describe and understand the observed Scottish population attendance risk.

Table 5 reports the modeled rate for every level of each factor. Consistent with Age being identified as the individual factor with the greatest improvement on the model in Section 5.1, modeled rates varied most widely across age bands, from 18 per 1,000 for 40-64 to 35 per 1,000 for 75 plus. Modeled rates for Health Boards showed a similarly wide range from 17 per 1,000 in S08000020 (NHS Grampian) to 32 in S08000025 (NHS Orkney), consistent with HBT's identification as the second most improving factor in our model. Deprivation showed a similar but somewhat smaller range from 19 per 1,000 in Quintile 5 to 30 per 1,000 in Quintile 1. Sex showed the smallest range of the four factors (just a difference of 2 per 1,000 population), consistent with exploratory findings in Section 3 and Sex contributing the smallest change in model fit when removed. Together, these population averaged results provided further support to the relative rankings established earlier in Section 5.1 as well as early on in Section 3 exploratory analysis.

Factor	Level	Modeled Rate (Attendance per 1,000 Population)	Factor	Level	Modeled Rate (Attendance per 1,000 Population)
Age	Under 18	25	Health Board	S08000015	21
	18-24	25		S08000016	26
	25-39	22		S08000017	32
	40-64	18		S08000019	24
	65-74	21		S08000020	17
	75 plus	35		S08000022	19
Sex	Female	23		S08000024	28
	Male	25		S08000025	32
Deprivation	1	30		S08000026	29
	2	28		S08000028	31
	3	24		S08000029	21
	4	21		S08000030	19
	5	19		S08000031	28
				S08000032	25

Table 5: Population averaged modeled attendance rate by factor and level.

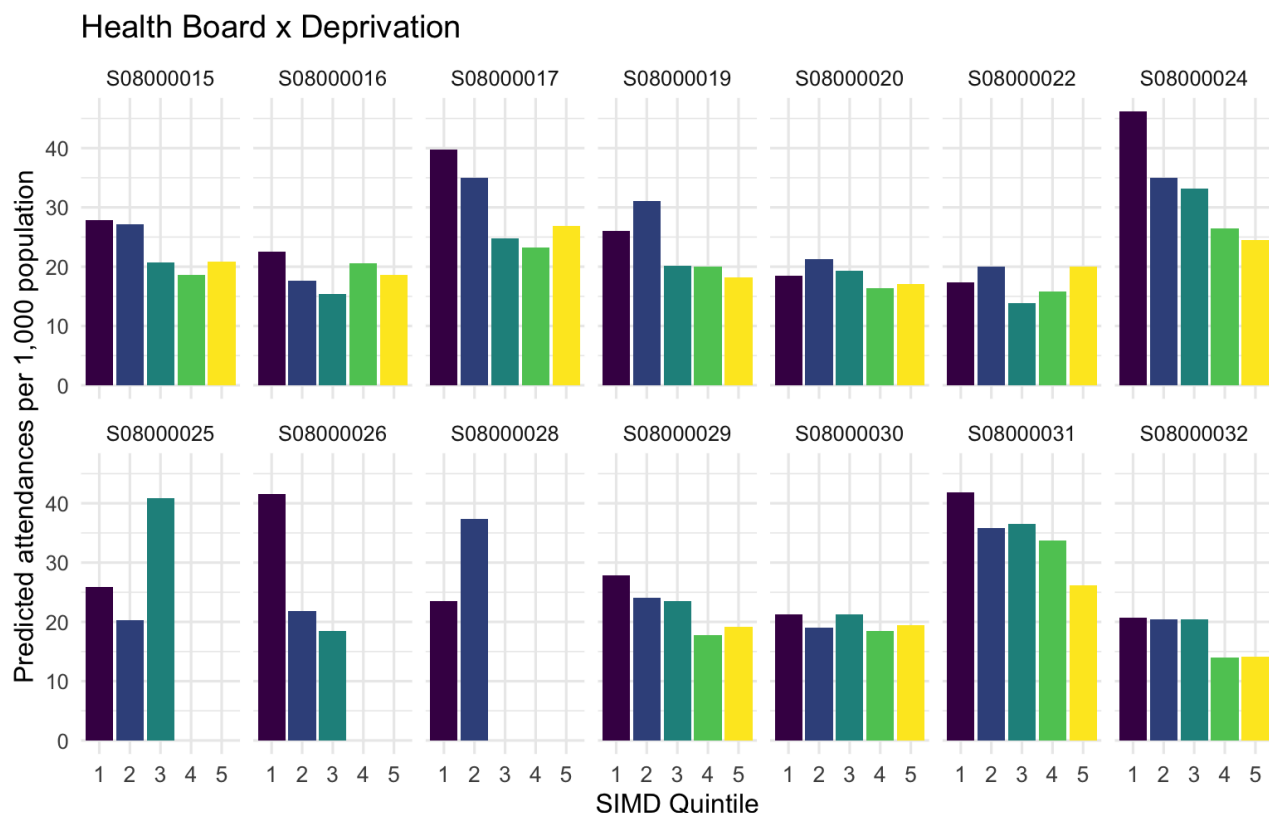


Figure 7: Modeled attendance rate across all five deprivation quintiles faceted by health board.

Modeled attendance rates were plotted for each of the six interactions, using fitted values from the chosen model. Figure 7 shows the modeled attendance rate across all five deprivation quintiles faceted by health board. Note that the gaps found in Quintiles 3, 4, and 5, for health boards S08000025 (NHS Orkney), S08000026 (NHS Shetland), and S08000028 (NHS Western Isles) are due to the non-existent resident population for those deprivation levels in those health boards. It is clear that the modeled attendance rate patterns found across deprivation quintiles differs between health boards. For instance, S08000024 (NHS Lothian) and S08000017 (NHS Dumfries and Galloway) saw a much more consistent and steep decrease in attendance rates from Quintile 1 to Quintile 5 whereas S08000030 (NHS Tayside) and S08000020 (NHS Grampian) saw much more flat measures of attendance rate across the quintiles. In general, Quintile 1 (most deprived) most often had the highest attendance rate whereas Quintile 5 generally had the lowest (least deprived).

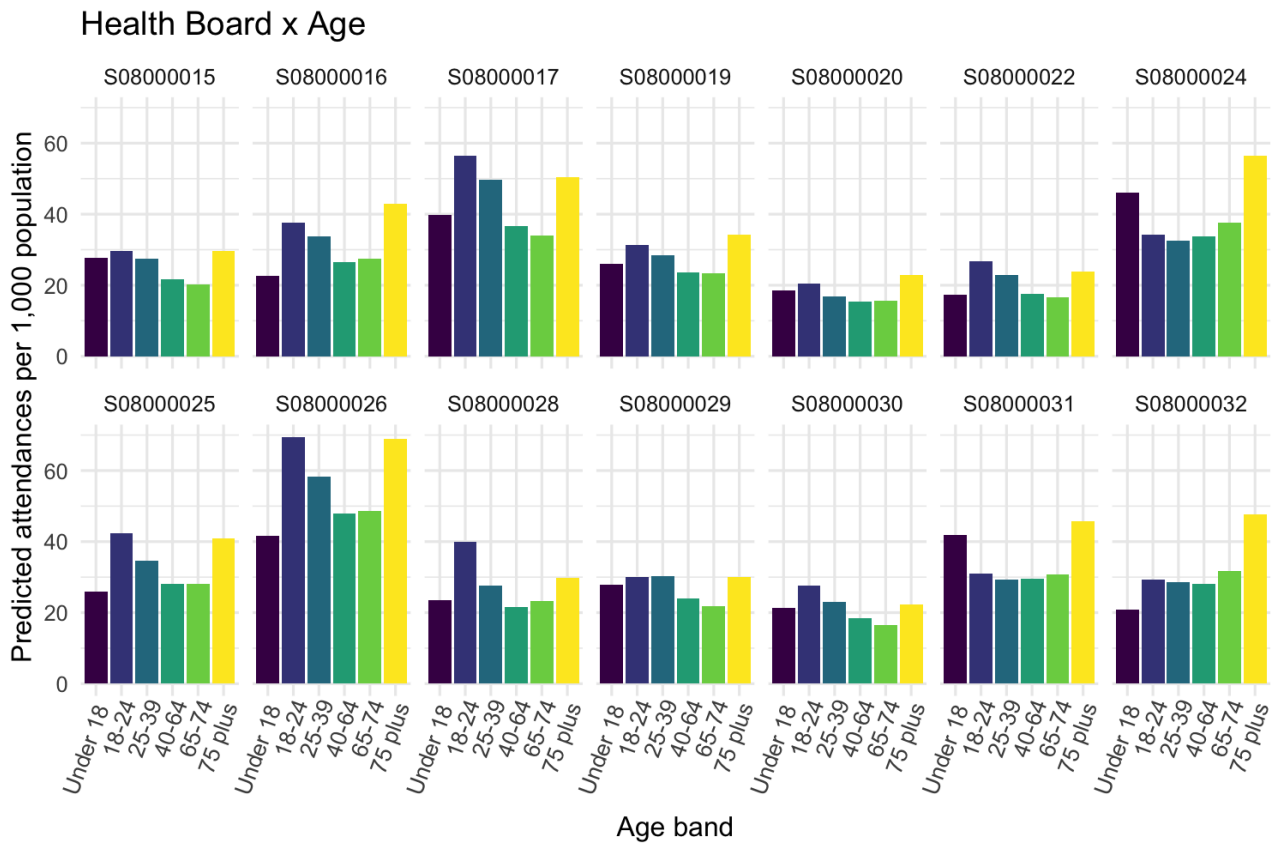


Figure 8: Modeled attendance rate across all six age groups faceted by health board.

Figure 8 shows the modeled attendance rate across all six age bands faceted by health board. Note Health Boards S08000024 (NHS Lothian) and S08000026 (NHS Shetland) saw particularly steep differences in modeled attendance rates between age bands compared to the more flat pattern found in S08000020 (NHS Grampian) and S08000029 (NHS Fife). In general, those Under 18, 18-24, and 75 plus saw the highest modeled attendance rates throughout all health boards.

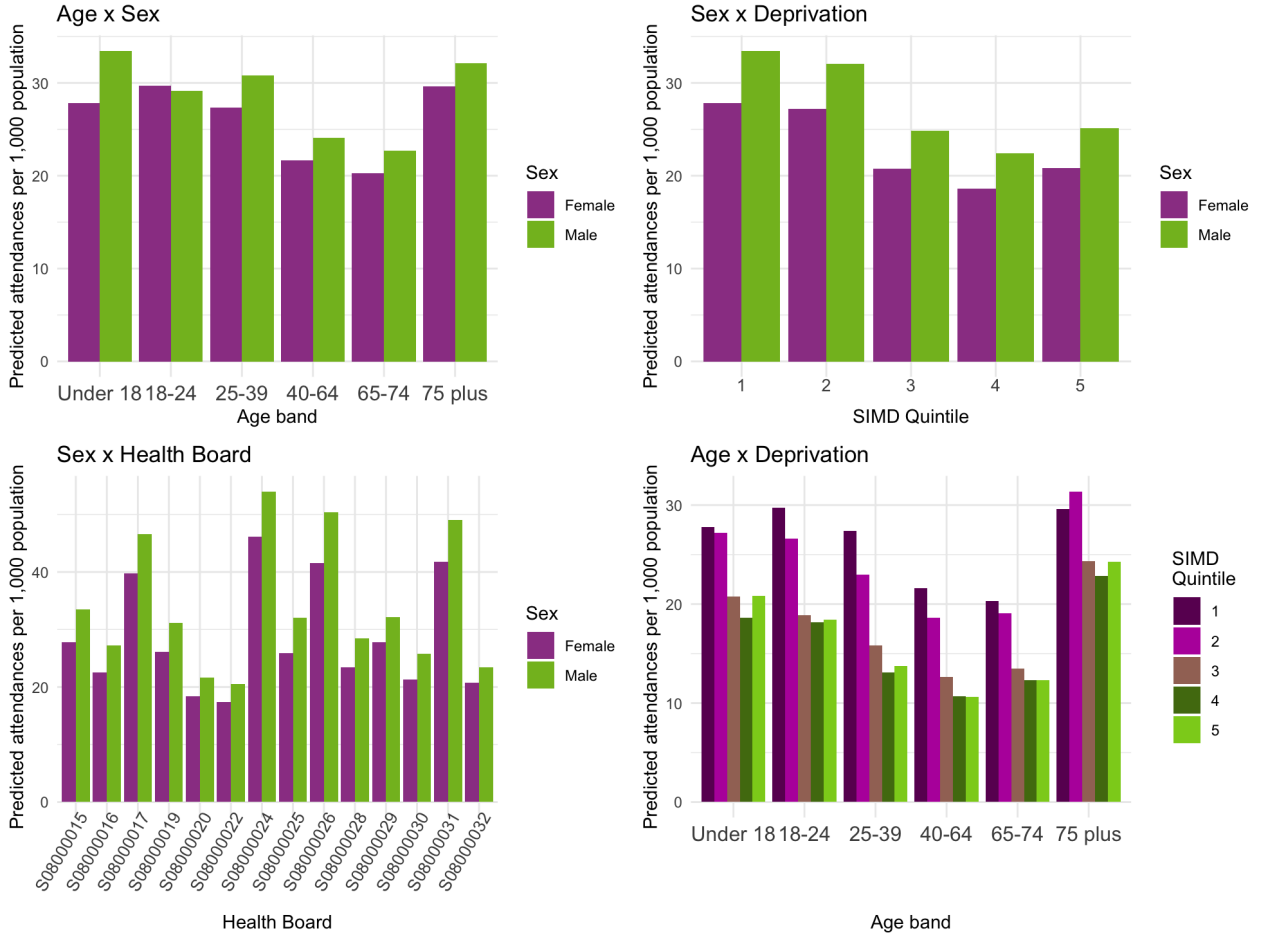


Figure 9: Modeled attendance rate for Age $\times$ Sex, Sex $\times$ Deprivation, Age $\times$ Deprivation, and Sex $\times$ HBT interactions.

Figure 9 presents the modeled attendance rate for the remaining interactions of Age $\times$ Sex, Sex $\times$ Deprivation, Age $\times$ Deprivation, and Sex $\times$ HBT. The gap between Male and Female attendance rates is relatively constant across age bands, deprivation quintiles, and health boards. Most deviation was found across the health boards such as S08000024 (NHS Lothian) and S08000026 (NHS Shetland) which saw much larger gaps between sex than S08000022 (NHS Highland) and S08000020 (NHS Grampian). These all correlate with earlier findings of sex being the weakest factor along with its interactions and showing little evidence of elevated risk in A&E attendance between Males and Females. The spread between deprivation quintiles in each age band was mostly consistent with the highest attendance rates most often occurring in Quintile 1 (most deprived) and the lower attendance rates occurring in Quintile 5 (least deprived) throughout each age group. The widest spread was seen in age band 25-39 and 40-64 suggesting that the deprivation pattern in risk of A&E attendance is concentrated in young adults and middle age.

Overall, the results from factor and interaction term removal comparisons made in Section 5.1 and the modeled attendance rates indicate that A&E attendance risk doesn't come from a single demographic characteristic. The highest population average modeled rates were observed for those 75 plus, those in Quintile 1 (most deprived), and those in S08000025 (NHS Orkney), S08000017 (NHS Dumfries and Galloway), and S08000028 (NHS Western Isles). Importantly, it was found that these factors did not act independently. For instance, while S08000024 (NHS Lothian) did not have the highest modeled attendance rate (28 per 1,000 population), it showed one of the steepest decreasing patterns from deprivation Quintile 1 to 5. Similarly, it also showed one of the steepest increases in attendance risk by age group. This tells us the combination of older age (75 plus) and higher deprivation (Quintile 1) carries elevated risk that could not be predicted by each factor alone. Stronger deprivation patterns

were found in those of adults (25-64) and comparatively flat for the oldest and youngest age groups suggesting that the contribution deprivation has on attendance risk is perhaps concentrated in certain stages of life. Sex contributed a stable but very small amount with little evidence of strong elevated attendance risk between Males and Females in any board or deprivation or age group. These findings tell us that the population at most risk for A&E attendance is not strictly defined by a single demographic factor or geography, rather mostly by a specific combination of age, deprivation level, and health board of treatment.

5.3 Subpopulation Risk

To get a picture of a few specific combinations of Age, Sex, Deprivation, and Health Board of particular interest, Table 6 shows ten subpopulations with the highest modeled A&E attendance rates and ten subpopulations with the lowest modeled A&E attendance rates.

Top 10, Highest Risk Subpopulations					Bottom 10, Lowest Risk Subpopulations				
Age	Sex	Deprivation	HBT	Rate	Age	Sex	Deprivation	HBT	Rate
75 plus	Male	3	S08000025	77	40-64	Female	5	S08000020	8
75 plus	Female	3	S08000025	71	40-64	Male	5	S08000020	8
18-24	Female	2	S08000028	62	25-39	Female	5	S08000020	9
75 plus	Male	4	S08000026	59	18-24	Female	5	S08000029	10
18-24	Female	2	S08000026	59	40-64	Female	4	S08000020	10
18-24	Male	3	S08000025	59	40-64	Female	5	S08000030	10
18-24	Female	1	S08000017	58	40-64	Female	3	S08000022	10
25-39	Male	1	S08000017	58	40-64	Male	4	S08000020	10
75 plus	Male	1	S08000024	57	25-39	Female	4	S08000020	11
18-24	Male	2	S08000028	57	40-64	Female	5	S08000015	11

Table 6: Ten subpopulation with the highest and ten with the lowest modeled attendance rates.

The ten high risk subpopulations were concentrated around deprivation Quintile 3 and 1 (60% of the 10 subpopulations) and those aged 75 plus and 18-24. There was some variance in health boards of treatment with highlights of NHS Orkney (S08000025), NHS Shetland (S08000026), NHS Western Isles (S08000028), and NHS Dumfries and Galloway (S08000017). The ten low risk subpopulations showed an opposite pattern with a concentration in deprivation Quintile 5, those aged 40-64, and those being treated at health board NHS Grampian (S08000020). An interesting note is that of the 10 subpopulations with the lowest risk, 80% were Female whereas the higher risk subpopulations saw a more even spread of sex with 40% Female. These types of findings have direct and applicable targeting for resource allocation and interventions as discussed in Section 6.

6 Discussion

6.1 Key Findings

Our research found that the populations at greatest risk of A&E attendance in Scotland are not strictly defined by single demographic characteristics rather by specific combinations of those demographics. For instance, when looking at the subpopulation with the highest modeled attendance rate (75 plus, Male, Quintile 3, S08000025) versus the subpopulation with the lowest modeled attendance rate (40-64, Female, Quintile 5, S08000020), it indicates that attendance risk can actually vary up to nearly 10 fold between groups of people in Scotland. In general, A&E attendance risk in Scotland tends to concentrate among the young (under 18 to 24), the older (75 plus), and the more deprived residents and this risk builds among certain health boards of treatment.

The increased risk among those aged 75 plus may correlate with their higher prevalence of multi morbidity. Previous research of NHS Lothian found that not only did about 10% of A&E attendees have multi morbidity but also that older individuals were more likely to have multi morbidity [1]. The high risk in the younger age bands is more likely to be related to acute unplanned visits [10]. For

instance, unplanned pediatric illnesses may cause guardians to use A&E as a first course of action if outside of primary care hours [13]. Interestingly, previous research has also found that young adults are more likely to use A&E for healthcare that could otherwise be provided in a “less urgent setting” and may also make use of emergency care for reasons such as convenience or lack of confidence in health care related decisions [10, 7].

A particularly stand out finding was the concentration of the higher attendance risk of those classified as more deprived and in particular, those of working age (aged 25-64). More deprived populations have previously been found to more often be in poorer health and have a higher mortality rate [3]. Accessibility of A&E stands out as a potential insight into why certain subpopulations are at higher risk of attendance. Previous research from England found that some of the deprivation pattern found in A&E attendance reflects the difficulty to access less “extreme” alternatives to A&E for treatment such as primary care GP rather than just differences in the medical help needed [8]. Additionally, other research has found that rural areas have a lower GP supply and can thus not keep up with demand, less permanent providers, and the increased challenge of potential extensive travel [4]. In Scotland, it was found that a proportion of A&E attendances that could be avoided are a result of the convenience and shorter wait times associated with A&E [6]. This relates to the rural health boards such as S08000025 (NHS Orkney) and S08000026 (NHS Shetland) which repeatedly showed up in the analysis and the modeled attendance rates as being of higher risk. The young working age population may find the convenience of A&E walk-in access with no appointment required easy compared to booking with a GP. Further research can be applied to this area to see if the effect of deprivation or geographic location on attendance may be influenced by access to alternative care. Additionally, if there was an improvement to primary care access and or added community care in these areas, could this help reduce avoidable A&E demand that is putting pressure on the system?

Our research and analysis consistently did not find sex to have a particularly strong effect on A&E attendance rate.

All together, these findings suggest that a uniform country wide strategy for treating A&E demand is unlikely to be effective.

6.2 Future Research

Several extensions can follow from this research. Given the distinct and strong effect that deprivation has on attendance risk, future research can be applied to see how decompositions of SIMD quintiles through indicators like alcohol/drug use, employment, and access to services, may drive this risk. This could have direct implication for targeting interventions towards the high deprivation population that make up A&E attendances. Further, frequent A&E attendance, defined as attending more than five times in a year, has been found to place a large financial and system resource burden [2]. This would be a natural next analysis using the same demographic framework in our models to understand not just who attends frequently and are thus placing a burden on the system, but also if there is a difference between the demographic makeup of those who attend generally versus frequently. This again would be a good analysis to understand where to target interventions. Additionally, the preliminary evidence of a shift in attendance patterns following the COVID-19 pandemic in Section 2.3 introduces an area that would benefit from further investigation. It would be insightful to see how attendance rates differ between the demographic subpopulations pre-pandemic and post-pandemic. And did the return to pre-pandemic rates see the same growth across subpopulations? This also introduces the idea of potentially including a time or seasonal component into models.

6.3 Limitations

Several limitations arise from the data and throughout our research. As previously stated, population estimates were only available up to 2024, where the remaining months of 2025 and 2026 used those same estimates. Ideally, we would have up to date population estimates. Health Board of Treatment is

where the patient was treated but may not match their Health Board of residence. This could potentially lead to false generalizations of certain health boards and patient attendances. The demographic data set provided by Public Health Scotland only included four demographic factors, Age, Sex, Deprivation, and Health Board. This does not account for other patient characteristics such as ethnicity, proximity to health services, or other specific SIMD indicators that may influence attendance risk.

A greater limitation comes from how different PHS datasets are published at different levels of granularity and this mismatch in some cases prevented certain datasets from being merged at all. For example, while attendance counts were available broken down by demographics, the SIMD indicators dataset which would have provided more detail into individuals visiting A&E (such as education, employment, and access to health services) was published at data zone level only. This limited the depth to which certain secondary questions could be pursued within the scope of this research. Additionally, attendance counts used in the monthly attendance dataset reflects attendance events instead of unique individuals, that is a person visiting multiple times is counted for each of their attendances. This can effect interpretations of the analyses if for instance there are frequent attenders bumping up attendance in certain subpopulations.

7 Conclusion

Within this report, we investigated how demographic factors influence A&E attendance risk in Scotland. We found that attendance risk is not explainable by a single factor, but strongly by the combination of age, deprivation, and health board. This was established using a regression setup applied to Scotland's A&E monthly attendance data. Given the pressures A&E faces in their resources, staff, and finances, these findings offer Public Health Scotland and individual health boards an understanding of where they may look into targeting certain resources or alternative services dependent on the subpopulations most at risk rather than applying a uniform plan nation wide. While this research was somewhat limited by the available public data, it provides a strong base that is reproducible and can be built on top of to apply to future research in the area. If we want to reduce preventable pressure on Scotland's health care system, in particular in the A&E department, we cannot treat the population as a single group but understanding that attendance risk is concentrated among specific demographic combinations of individuals.

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Appendices

A Health Board Name and Country Code

HBT Code	HBT Name
S08000015	NHS Ayrshire and Arran
S08000016	NHS Borders
S08000017	NHS Dumfries and Galloway
S08000019	NHS Forth Valley
S08000020	NHS Grampian
S08000022	NHS Highland
S08000024	NHS Lothian
S08000025	NHS Orkney
S08000026	NHS Shetland
S08000028	NHS Western Isles
S08000029	NHS Fife
S08000030	NHS Tayside
S08000031	NHS Greater Glasgow and Clyde
S08000032	NHS Lanarkshire

Table 7: NHS Scotland Health Board geography codes and corresponding names.

B Model Coefficients

Term	Estimate (β)	Standard Error	$\exp(\beta)$ (Incidence Rate Ratio)	P-value
(Intercept)	-3.58281	0.008973	0.027797	<2e-16
Age18-24	0.066545	0.011362	1.068809	0
Age25-39	-0.01538	0.010479	0.984736	0.142134
Age40-64	-0.25106	0.010087	0.777976	<2e-16
Age65-74	-0.31419	0.010963	0.730379	<2e-16
Age75 plus	0.063984	0.010604	1.066075	0
SexMale	0.185645	0.006874	1.203995	<2e-16
Deprivation2	-0.02218	0.009556	0.978065	0.020291
Deprivation3	-0.29294	0.009836	0.746067	<2e-16
Deprivation4	-0.40221	0.010175	0.66884	<2e-16
Deprivation5	-0.28706	0.010441	0.750463	<2e-16
HBTS08000016	-0.21039	0.015596	0.810267	<2e-16
HBTS08000017	0.357691	0.013015	1.430024	<2e-16
HBTS08000019	-0.06349	0.011732	0.938481	0
HBTS08000020	-0.41118	0.012373	0.662869	<2e-16
HBTS08000022	-0.47118	0.012672	0.624266	<2e-16
HBTS08000024	0.507407	0.010948	1.660979	<2e-16
HBTS08000025	-0.06981	0.019953	0.93257	0.000467
HBTS08000026	0.401706	0.018831	1.494372	<2e-16
HBTS08000028	-0.16852	0.019893	0.844916	<2e-16
HBTS08000029	0.000119	0.01151	1.000119	0.991721
HBTS08000030	-0.26723	0.011598	0.765493	<2e-16

Table 8 continued from previous page

Term	Estimate (β)	Standard Error	$\exp(\beta)$ (Incidence Rate Ratio)	P-value
HBTS08000031	0.408458	0.010776	1.504496	<2e-16
HBTS08000032	-0.29135	0.011112	0.747254	<2e-16
Age18-24:SexMale	-0.2041	0.005542	0.815377	<2e-16
Age25-39:SexMale	-0.06833	0.005156	0.933956	<2e-16
Age40-64:SexMale	-0.07732	0.004967	0.925593	<2e-16
Age65-74:SexMale	-0.07485	0.005387	0.927879	<2e-16
Age75 plus:SexMale	-0.10574	0.005181	0.89966	<2e-16
Age18-24:Deprivation2	-0.08841	0.008662	0.915388	<2e-16
Age25-39:Deprivation2	-0.15234	0.008066	0.858692	<2e-16
Age40-64:Deprivation2	-0.12604	0.007911	0.881575	<2e-16
Age65-74:Deprivation2	-0.04057	0.008666	0.960244	0.000003
Age75 plus:Deprivation2	0.07967	0.00847	1.082929	<2e-16
Age18-24:Deprivation3	-0.16156	0.008745	0.850817	<2e-16
Age25-39:Deprivation3	-0.25649	0.008102	0.773763	<2e-16
Age40-64:Deprivation3	-0.24263	0.007894	0.784565	<2e-16
Age65-74:Deprivation3	-0.11541	0.00865	0.890998	<2e-16
Age75 plus:Deprivation3	0.095669	0.00843	1.100395	<2e-16
Age18-24:Deprivation4	-0.09035	0.008905	0.913608	<2e-16
Age25-39:Deprivation4	-0.33287	0.008269	0.716863	<2e-16
Age40-64:Deprivation4	-0.29946	0.008008	0.741219	<2e-16
Age65-74:Deprivation4	-0.09704	0.008766	0.907516	<2e-16
Age75 plus:Deprivation4	0.143017	0.008515	1.153749	<2e-16
Age18-24:Deprivation5	-0.1918	0.009216	0.825472	<2e-16
Age25-39:Deprivation5	-0.40141	0.008669	0.669377	<2e-16
Age40-64:Deprivation5	-0.42134	0.008326	0.656167	<2e-16
Age65-74:Deprivation5	-0.21088	0.009123	0.809872	<2e-16
Age75 plus:Deprivation5	0.087868	0.008797	1.091843	<2e-16
Age18-24:HBTS08000016	0.443962	0.01701	1.558871	<2e-16
Age25-39:HBTS08000016	0.421081	0.01509	1.523608	<2e-16
Age40-64:HBTS08000016	0.411201	0.01406	1.508628	<2e-16
Age65-74:HBTS08000016	0.512829	0.015489	1.670008	<2e-16
Age75 plus:HBTS08000016	0.579549	0.014565	1.785233	<2e-16
Age18-24:HBTS08000017	0.28459	0.015117	1.329217	<2e-16
Age25-39:HBTS08000017	0.238687	0.013535	1.269581	<2e-16
Age40-64:HBTS08000017	0.170507	0.012848	1.185906	<2e-16
Age65-74:HBTS08000017	0.155055	0.014168	1.167723	<2e-16
Age75 plus:HBTS08000017	0.175287	0.013427	1.191589	<2e-16
Age18-24:HBTS08000019	0.117858	0.013649	1.125084	<2e-16
Age25-39:HBTS08000019	0.104735	0.012539	1.110417	<2e-16
Age40-64:HBTS08000019	0.155122	0.011976	1.1678	<2e-16
Age65-74:HBTS08000019	0.206989	0.01315	1.229969	<2e-16
Age75 plus:HBTS08000019	0.208241	0.012581	1.23151	<2e-16
Age18-24:HBTS08000020	0.036537	0.013396	1.037213	0.006381
Age25-39:HBTS08000020	-0.07113	0.012342	0.931341	0
Age40-64:HBTS08000020	0.064025	0.011842	1.066119	0
Age65-74:HBTS08000020	0.142814	0.012914	1.153516	<2e-16
Age75 plus:HBTS08000020	0.156471	0.01239	1.169377	<2e-16
Age18-24:HBTS08000022	0.365133	0.014328	1.440706	<2e-16
Age25-39:HBTS08000022	0.288452	0.01303	1.33436	<2e-16

Table 8 continued from previous page

Term	Estimate (β)	Standard Error	$\exp(\beta)$ (Incidence Rate Ratio)	P-value
Age40-64:HBTS08000022	0.258589	0.012425	1.295102	<2e-16
Age65-74:HBTS08000022	0.269335	0.013606	1.309093	<2e-16
Age75 plus:HBTS08000022	0.254851	0.013017	1.290269	<2e-16
Age18-24:HBTS08000024	-0.36774	0.012598	0.692298	<2e-16
Age25-39:HBTS08000024	-0.33588	0.011729	0.714706	<2e-16
Age40-64:HBTS08000024	-0.06158	0.01132	0.940275	0
Age65-74:HBTS08000024	0.10887	0.012099	1.115017	<2e-16
Age75 plus:HBTS08000024	0.138393	0.011701	1.148426	<2e-16
Age18-24:HBTS08000025	0.42601	0.028685	1.531136	<2e-16
Age25-39:HBTS08000025	0.30203	0.023722	1.352602	<2e-16
Age40-64:HBTS08000025	0.329248	0.021556	1.389922	<2e-16
Age65-74:HBTS08000025	0.391873	0.024902	1.47975	<2e-16
Age75 plus:HBTS08000025	0.39296	0.022581	1.481359	<2e-16
Age18-24:HBTS08000026	0.447957	0.026773	1.565111	<2e-16
Age25-39:HBTS08000026	0.354216	0.022643	1.425063	<2e-16
Age40-64:HBTS08000026	0.394203	0.021003	1.483202	<2e-16
Age65-74:HBTS08000026	0.471169	0.024598	1.601866	<2e-16
Age75 plus:HBTS08000026	0.441328	0.022486	1.554771	<2e-16
Age18-24:HBTS08000028	0.464071	0.028768	1.590535	<2e-16
Age25-39:HBTS08000028	0.179486	0.025149	1.196602	0
Age40-64:HBTS08000028	0.167542	0.0225	1.182395	0
Age65-74:HBTS08000028	0.309664	0.025359	1.362967	<2e-16
Age75 plus:HBTS08000028	0.175936	0.024055	1.192362	0
Age18-24:HBTS08000029	0.00745	0.013528	1.007478	0.581832
Age25-39:HBTS08000029	0.102598	0.01241	1.108045	<2e-16
Age40-64:HBTS08000029	0.102314	0.011877	1.107731	<2e-16
Age65-74:HBTS08000029	0.06753	0.012993	1.069862	0
Age75 plus:HBTS08000029	0.016031	0.012458	1.01616	0.198143
Age18-24:HBTS08000030	0.19257	0.013331	1.212362	<2e-16
Age25-39:HBTS08000030	0.096018	0.012382	1.100779	0
Age40-64:HBTS08000030	0.110651	0.011886	1.117005	<2e-16
Age65-74:HBTS08000030	0.052938	0.013027	1.054365	0.000048
Age75 plus:HBTS08000030	-0.01424	0.012476	0.985864	0.253825
Age18-24:HBTS08000031	-0.36906	0.012477	0.691384	<2e-16
Age25-39:HBTS08000031	-0.33928	0.01167	0.712281	<2e-16
Age40-64:HBTS08000031	-0.09768	0.01127	0.90694	<2e-16
Age65-74:HBTS08000031	0.007239	0.011983	1.007266	0.545751
Age75 plus:HBTS08000031	0.027528	0.011614	1.02791	0.017775
Age18-24:HBTS08000032	0.279952	0.013037	1.323066	<2e-16
Age25-39:HBTS08000032	0.338459	0.012048	1.402784	<2e-16
Age40-64:HBTS08000032	0.549871	0.011557	1.73303	<2e-16
Age65-74:HBTS08000032	0.736869	0.01232	2.089383	<2e-16
Age75 plus:HBTS08000032	0.76567	0.011938	2.150435	<2e-16
SexMale:Deprivation2	-0.02156	0.004879	0.978672	0.000010
SexMale:Deprivation3	-0.00539	0.004888	0.99462	0.269729
SexMale:Deprivation4	0.001468	0.004967	1.001469	0.767648
SexMale:Deprivation5	0.001361	0.005181	1.001361	0.792856
SexMale:HBTS08000016	0.004152	0.008775	1.004161	0.636060
SexMale:HBTS08000017	-0.02654	0.008086	0.973806	0.001028

Table 8 continued from previous page

Term	Estimate (β)	Standard Error	$\exp(\beta)$ (Incidence Rate Ratio)	P-value
SexMale:HBTS08000019	-0.0096	0.007514	0.990448	0.201496
SexMale:HBTS08000020	-0.02304	0.007402	0.977221	0.001851
SexMale:HBTS08000022	-0.01789	0.00776	0.982269	0.021146
SexMale:HBTS08000024	-0.02866	0.006985	0.971749	0.000041
SexMale:HBTS08000025	0.025879	0.013802	1.026217	0.060794
SexMale:HBTS08000026	0.006506	0.013573	1.006527	0.631695
SexMale:HBTS08000028	0.006166	0.014499	1.006185	0.670653
SexMale:HBTS08000029	-0.04063	0.007447	0.960185	0.000000
SexMale:HBTS08000030	0.008109	0.00742	1.008142	0.274469
SexMale:HBTS08000031	-0.02701	0.006926	0.973348	0.000096
SexMale:HBTS08000032	-0.0659	0.007116	0.936229	<2e-16
Deprivation2:HBTS08000016	-0.22348	0.014923	0.799733	<2e-16
Deprivation3:HBTS08000016	-0.08615	0.014514	0.917455	0
Deprivation4:HBTS08000016	0.308413	0.014796	1.361264	<2e-16
Deprivation5:HBTS08000016	0.099039	0.016857	1.104109	0
Deprivation2:HBTS08000017	-0.10685	0.012121	0.898657	<2e-16
Deprivation3:HBTS08000017	-0.18271	0.012402	0.833007	<2e-16
Deprivation4:HBTS08000017	-0.137	0.01335	0.871971	<2e-16
Deprivation5:HBTS08000017	-0.10304	0.014292	0.90209	0
Deprivation2:HBTS08000019	0.198145	0.011146	1.21914	<2e-16
Deprivation3:HBTS08000019	0.038162	0.011646	1.0389	0.001049
Deprivation4:HBTS08000019	0.137505	0.011881	1.147407	<2e-16
Deprivation5:HBTS08000019	-0.07344	0.012094	0.929193	0
Deprivation2:HBTS08000020	0.166362	0.012	1.181001	<2e-16
Deprivation3:HBTS08000020	0.343418	0.012164	1.409758	<2e-16
Deprivation4:HBTS08000020	0.280269	0.01235	1.323486	<2e-16
Deprivation5:HBTS08000020	0.208799	0.012505	1.232197	<2e-16
Deprivation2:HBTS08000022	0.162565	0.012056	1.176524	<2e-16
Deprivation3:HBTS08000022	0.063869	0.012113	1.065952	0
Deprivation4:HBTS08000022	0.312807	0.01239	1.367258	<2e-16
Deprivation5:HBTS08000022	0.427961	0.013473	1.534127	<2e-16
Deprivation2:HBTS08000024	-0.256	0.010462	0.774144	<2e-16
Deprivation3:HBTS08000024	-0.03592	0.010798	0.964721	0.000880
Deprivation4:HBTS08000024	-0.15244	0.011078	0.858609	<2e-16
Deprivation5:HBTS08000024	-0.34496	0.011162	0.708246	<2e-16
Deprivation2:HBTS08000025	-0.22278	0.019481	0.800291	<2e-16
Deprivation3:HBTS08000025	0.748018	0.016662	2.112808	<2e-16
Deprivation4:HBTS08000025	NA	NA	NA	NA
Deprivation5:HBTS08000025	NA	NA	NA	NA
Deprivation2:HBTS08000026	-0.62173	0.025865	0.537017	<2e-16
Deprivation3:HBTS08000026	-0.518	0.015948	0.595709	<2e-16
Deprivation4:HBTS08000026	NA	NA	NA	NA
Deprivation5:HBTS08000026	NA	NA	NA	NA
Deprivation2:HBTS08000028	0.486512	0.016285	1.626633	<2e-16
Deprivation3:HBTS08000028	NA	NA	NA	NA
Deprivation4:HBTS08000028	NA	NA	NA	NA
Deprivation5:HBTS08000028	NA	NA	NA	NA
Deprivation2:HBTS08000029	-0.12039	0.011006	0.886571	<2e-16
Deprivation3:HBTS08000029	0.122207	0.011369	1.129988	<2e-16

Table 8 continued from previous page

Term	Estimate (β)	Standard Error	$\exp(\beta)$ (Incidence Rate Ratio)	P-value
Deprivation4:HBTS08000029	-0.04867	0.01179	0.952492	0.000037
Deprivation5:HBTS08000029	-0.08241	0.011877	0.920894	0
Deprivation2:HBTS08000030	-0.08694	0.011247	0.916734	0
Deprivation3:HBTS08000030	0.291036	0.011397	1.337813	<2e-16
Deprivation4:HBTS08000030	0.259265	0.011596	1.295977	<2e-16
Deprivation5:HBTS08000030	0.195786	0.011976	1.216266	<2e-16
Deprivation2:HBTS08000031	-0.13118	0.010264	0.87706	<2e-16
Deprivation3:HBTS08000031	0.156591	0.0106	1.169517	<2e-16
Deprivation4:HBTS08000031	0.186514	0.010881	1.205042	<2e-16
Deprivation5:HBTS08000031	-0.18205	0.011017	0.833563	<2e-16
Deprivation2:HBTS08000032	0.008499	0.010419	1.008535	0.414679
Deprivation3:HBTS08000032	0.277773	0.010811	1.320186	<2e-16
Deprivation4:HBTS08000032	0.004669	0.011185	1.00468	0.676351
Deprivation5:HBTS08000032	-0.10204	0.011521	0.902989	<2e-16

Table 8: Full coefficient table for the final population attendance rate model, including Incidence Rate Ratios ($\exp(\text{Estimate})$). Coefficients with NA could not be estimated due to zero observed population.